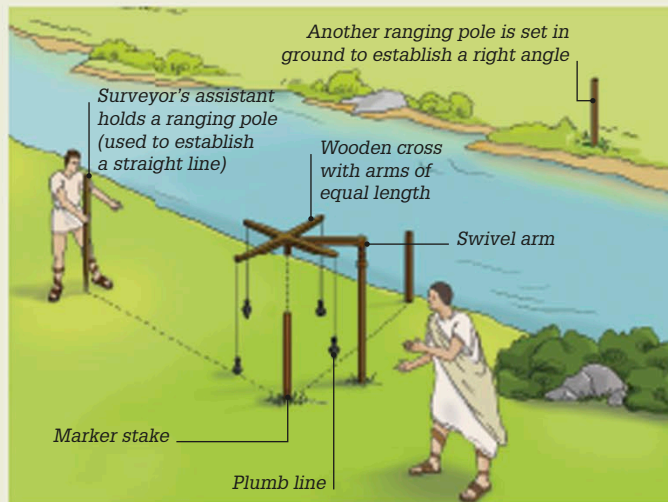




Scale marked
in degrees

**Gunner's caliper
from Venice, Italy,
16th century**



Surveyor's tool

The Romans, who were skilled builders and surveyors, used a device called a groma to measure right angles. It consisted of a horizontal wooden cross and a weighted cord (plumb line), which hung from each of the four arms. The surveyor would look down each pair of plumb lines in turn to establish a right angle.

Metric and imperial systems

The metric system, first introduced in France, is the official system of measurement in most countries today. The imperial system was once used throughout the British Empire. The US is the only major country that still uses it officially—it is known there as the customary measurement system. While both systems are still in use, the values are not directly equivalent.

Metric	Imperial
Centimeter (cm)	Inch (in)
Meter (m)	Foot (ft)
Kilometer (km)	Mile (m)
Kilometer per hour (km/h)	Mile per hour (mph)
Gram (g)	Ounce (oz)
Kilogram (kg)	Pound (lb)
Liter (l)	Gallon (gal)
Celsius (°C)	Fahrenheit (°F)



Sailor's tobacco box

Dual-purpose device

The lid of this small tobacco box, which possibly belonged to an 18th-century seaman, was engraved with a perpetual calendar (meaning it is valid for numerous years) so he always knew what day of the week it was. Mathematical tables on the bottom, used with the ship's log float (see p.93), also enabled him to calculate the ship's speed.

Perpetual calendar
engraved on lid

1875

The Treaty of the Meter, signed in Paris by representatives of 17 nations, agreed to international standard units of measurement based on the meter and the kilogram.

1960

The International System of Units (SI), the modern form of the metric system, was officially adopted. It is the most widely used system of measurement.

**Leica DISTO D3
is an LDM**



1993

The handheld Laser Distance Meter (LDM) came into use. It shoots a laser pulse at a distant object and measures the time taken for the pulse to be reflected back.